

standard MAL-BIND-K- $\{CT, PK\}$ + syntactic K-CTX extension
CDM Fig. 6 for CT/PK; K-CTX outside the CDM lattice; both rejection styles
machine-checked in EasyCrypt — **zero KEM binding assumption**

reduces to
(no KEM property used)

KEM-aware game (MAL adversary supplies the keypairs)
 $K = H(\text{encode}[ss_{pq}, ss_{tr}, ct_{pq}, pk_{pq}, ct_{tr}, pk_{tr}, ctx])$
via Decaps + combiner
implicit reject: $\text{malbind}_{\{kct, kpk, kctx\}}\text{le_cr}$
explicit reject ($K \neq \perp$ load-bearing): $\text{malbind}_{*}\text{xrej_le_cr}$

a differing $ct/pk/ctx$
 $\Rightarrow$ *a differing field list*

encode_inj — injective length-prefixed encoding: **PROVED lemma**
 lp_cat_inj via eqseq_cat ; bottoms out at $\text{be8_size} + \text{be8_inj}$

differing encodings,
equal hash = a collision

CR(SHA3-256) — the *sole*
cryptographic assumption

Honest scope. $\text{decaps}/\text{accepts}$ are abstract & axiom-free (the result holds for *every* Decaps), so: no FIPS-203 linkage; the ss fields are *inert* in the argument (binding flows through $ct/pk/ctx$). CR(SHA3) is assumed; IND-CCA2 is on paper; no $\text{spec} \leftrightarrow \text{impl}$ linkage.